

Example 1

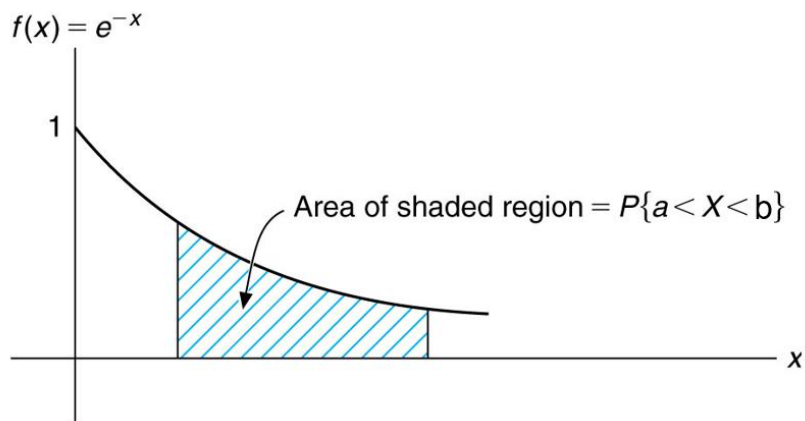
Consider the function

$$P(X = x) = \begin{cases} e^{-x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- a) Prove that it is a continuous probability distribution.
- b) Explain why it is incorrect to say that the “probability that $X = 0$ is 1”.

Answers

- a) $0 \leq P(X = x) \leq 1$. It fulfils Axiom 1 $0 \leq P(E) \leq 1$
 $\int_0^{\infty} e^{-x} dx = 1$. It fulfils Axiom 2 $P(S) = 1$
Axiom 3 (mutually exclusive events) are also fulfilled because of the definition of a function.
- b) For a continuous random variable, the probability of occurrence of a single point is 0.
For an interval (a, b) , the area subtended by the function gives the probability $P(a < X < b)$



Example 2

In a course, there are 44 male and 8 female students. Suppose they are ranked according to their scores in Quiz 1.

Assume that no two scores are the same and all rankings are equally likely.

Let X denote the highest ranking achieved by a female student. Find $P\{X = 3\}$.

Answers

The first and second ranked students are male. The third ranked student is female. The probability is

$$P\{X = 3\} = \left(\frac{44}{52}\right) \left(\frac{43}{51}\right) \left(\frac{8}{50}\right) = 0.114147813 \approx 11.4\%$$

Example 3

A pair of dice are rolled. Assume that the dice is fair. Calculate

- a) the expected value
 - b) the standard deviation
- of the sum of the two dice.

Answers

- a) Let X_1 and X_2 be the number obtained for the first and second die respectively.

$$E[X_1 + X_2] = E[X_1] + E[X_2] = 3.5 + 3.5 = 7$$

- b) Since we may assume that the number obtained for the first and the second die are independent,

$$Var[X_1 + X_2] = Var[X_1] + Var[X_2] = \frac{35}{12} + \frac{35}{12} = 5.833333333$$

$$\text{Standard deviation} = \sqrt{5.833333333} = 2.415229458 \approx 2.42$$

Example 4

In nature, there are some un-explained correlation phenomenon, for example, the EPR paradox (https://en.wikipedia.org/wiki/EPR_paradox). At other times, two random variables are correlated, but in very complicated ways that are difficult to model or analyze, for example, the trends of two stocks.

Consider two fair dice. Assume that the two die are correlated in a mysterious way with a covariance of 1.2. Calculate

- a) the expected value
 - b) the standard deviation
- of the sum of the two dice.

Answers

- a) Let X_1 and X_2 be the number obtained for the first and second die respectively. Linearity of expectation holds no matter what is the relationship between the two random variables. Hence

$$E[X_1 + X_2] = E[X_1] + E[X_2] = 3.5 + 3.5 = 7$$

- b) Since we may assume that the number obtained for the first and the second die are not independent,

$$Var[X_1 + X_2] = Var[X_1] + Var[X_2] + Cov(X_1, X_2) + Cov(X_2, X_1)$$

$$= \frac{35}{12} + \frac{35}{12} + 1.2 + 1.2 = 8.233333333$$

$$\text{Standard deviation} = \sqrt{8.233333333} = 2.869378562 \approx 2.87$$

Example 5

There are 151 different first generation Pokémon that can be caught in Hong Kong. Suppose a Pokémon appears at random,

- a) how many types of Pokémon will you expect to catch in 100 successful catches?
- b) how many catches is expected before all Pokémon is collected?

[Hint: See the Coupon Collector Problem

(https://en.wikipedia.org/wiki/Coupon_collector%27s_problem)]

- c) Discuss whether your estimate in b) is reasonable.

Answers

a)

Let X denote the number of different Pokémon in 100 successful catches.

$$X = X_1 + \cdots + X_{151}$$

$$X_i = \begin{cases} 1 & \text{if at least one type } i \text{ Pokémon is in the 100 successful catches} \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} E[X_i] &= 1P\{X_i = 1\} + 0P\{X_i = 0\} = P\{X_i = 1\} \\ &= P\{\text{at least one type } i \text{ Pokémon is in 100 successful catches}\} \\ &= 1 - P\{\text{no type } i \text{ Pokémon are in 100 successful catches}\} \\ &= 1 - \left(\frac{150}{151}\right)^{100} = 0.485445742 \end{aligned}$$

$$E[X] = E[X_1] + \cdots + E[X_{151}] = 151 \left[1 - \left(\frac{150}{151}\right)^{100} \right] = 73.30230709 \approx 73$$

b)

One cannot simply put $E[X] = 151$

$$151 \left[1 - \left(\frac{150}{151}\right)^n \right] = 151 \Rightarrow n = \infty$$

This is correct as to have expected value of 151, we need to have an infinite number of catches, as there is a very small chance we fail to catch a Pokémon after a large number of attempts.

We can find the expected number of attempts by the celebrated Coupon Collector's Theorem

The key to solving the problem is based on the observation that it takes very little time to collect the first few Pokémon, but it takes a long time to collect the last few Pokémon.

Let n be the number of different types of Pokémon. Let X be the number of catches until at least one of every type of Pokémon is caught.

Let X_i be the number of catches until a new Pokémon is caught while you have already collected $i - 1$ different types of Pokémon. When exactly $i - 1$ different types of Pokémon have been collected, the probability of obtaining a new type of Pokémon is

$$p_i = 1 - \frac{i - 1}{n}$$

Hence X_i is a geometric random variable with parameter p_i , and

$$E[X_i] = \frac{1}{p_i} = \frac{n}{n - i + 1}$$

Using linearity of expectations,

$$\begin{aligned} E[X] &= E\left[\sum_{i=1}^n X_i\right] \\ &= \sum_{i=1}^n E[X_i] \\ &= \sum_{i=1}^n \frac{n}{n - i + 1} \\ &= n \sum_{i=1}^n \frac{1}{i} \end{aligned}$$

$\sum_{i=1}^n \frac{1}{i}$ is known as the harmonic number $H(n)$.

In our question, $n = 151$.

Using the harmonic series calculator

(<https://www.math.utah.edu/~carlson/teaching/calculus/harmonic.html>)

[If needed, the harmonic series data will be given in tests and exams]

$$E[X] = (151)(5.597803105200172) \approx 846$$

c) It is known that some Pokémon can only be caught at certain districts in the world. For example, the Kangaskhan Pokémon can only be caught in areas near Australia. Thus the chance of catching a Pokémon is not equally likely.

Also, certain Pokémon only become available at a specific place at a specified period of time. Moreover, certain Pokémon only becomes catchable by invitation.

Example 6

We have 3 random variables X, Y, Z . Given that X and Z are independent. Simplify $Var(2X - Y + 4Z)$

Answers

As X and Z are independent

$$Var(2X - Y + 4Z)$$

$$= Var(2X) + Var(-Y) + Var(4Z) + Cov(2X, -Y) + Cov(-Y, 2X) + Cov(-Y, 4Z) + Cov(4Z, -Y)$$

Since

$$Cov(X, Y) = Cov(Y, X)$$

$$= Var(2X) + Var(-Y) + Var(4Z) + 2Cov(2X, -Y) + 2Cov(-Y, 4Z)$$

As

$$Var(aX) = a^2Var(X)$$

$$Cov(aX, bY) = abCov(X, Y)$$

$$= 4Var(X) + Var(Y) + 16Var(Z) - 4Cov(X, Y) - 8Cov(Y, Z)$$

Example 7

A random process (X, Y) for which X and Y are random variables records the following entries

$(1, 2) (2, 4) (3, 6) (4, 0)$

Compute the Covariance between the two variables.

Answers

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

$$E[XY] = \frac{2 + 8 + 18 + 0}{4} = 7$$

$$E[X] = \frac{1 + 2 + 3 + 4}{4} = 2.5$$

$$E[Y] = \frac{2 + 4 + 6 + 0}{4} = 3$$

$$\text{Cov}(X, Y) = 7 - 7.5 = -0.5$$

Example 8

Find the “expectation of life at birth by sex” from the Hong Kong Census and Statistics Department (<http://www.censtatd.gov.hk/hkstat/hkif/index.jsp>. See under “Health”)

Use the 2021 data below:

Number of births		Crude birth rate (per 1000 population)	Total fertility rate (per 1000 female population) (1) (2)	Number of deaths		Crude death rate (per 1000 population)	Expectation of life at birth		Number of marriages		Crude marriage rate (per 1000 population)	Median age at first marriage		Child dependency ratio (3)	Elderly dependency ratio (4)	Overall dependency ratio (5)
('000)	Year-on-year % change	Rate	Rate	('000)	Year-on-year % change	Rate	(years)		('000)	Year-on-year % change	Rate	(years)		Rate	Rate	Rate
Total	Total	Total	Total	Total	Total	Total	Male	Female	Total	Total	Total	Male	Female	Total	Total	Total
53.7	-5.0	7.2	1 080	47.4	+1.2	6.4	82.3	87.6	49.3	-4.8	6.6	31.5	29.7	159	238	397
52.9	-1.6	7.0	1 064	49.0	+3.3	6.5	82.4	88.1	44.2	-10.3	5.9	31.6	29.9	160	249	409
43.0	-18.6	5.8	883	50.7	+3.5	6.8	83.4	87.7	27.9	-37.0	3.7	31.9	30.4	159	262	421
37.0	-14.1	5.0	772	51.2 p	+1.0 p	6.9 p	83.0 p	87.7 p	26.9	-3.5	3.6	32.2	30.6	157	282	438

You may either use the male figure or the female figure in the question below.

- Provide an estimate of the probability that a person will live to 100 years old using Markov's inequality.
- If the standard deviation of the expectation of life at birth is 5 years, find the probability that a person will live to more than ± 20 years from the expected life at birth.
- Cantelli's inequality is a generalization of Chebyshev's inequality to one sided situations (https://en.wikipedia.org/wiki/Cantelli%27s_inequality). Assume the standard deviation of the expectation of life at birth is 5 years, estimate the probability that a person will live to 100 years old using Cantelli's inequality.
- Comment on your estimates. Also, suggest how you can estimate the expectation of life at birth 2 years later.
- Consider the population pyramid in Hong Kong (e.g. <https://www.populationpyramid.net/china-hong-kong-sar/2021/>)

Go back and forth in time to see how the pyramid is changing.

Also, reference the sex ratio at birth (https://en.wikipedia.org/wiki/Human_sex_ratio)

Suggest a reason why the sex ratio in Hong Kong is dropping.

Answers

a)

From the table, the expectation of life at birth by sex in 2021 is 83.0 for male and 87.7 for female. Using male data, $E[X] = 83.0$. Using Markov's inequality,

$$P\{X \geq 100\} \leq \frac{83.0}{100} = 0.83 = 83\%$$

Using female data,

$$P\{X \geq 100\} \leq \frac{87.7}{100} = 0.877 = 87.7\%$$

b)

$$P\{|X - E[X]| \geq 20\} \leq \frac{5^2}{20^2} \approx 6.25\%$$

c)

$$P\{X - E[X] \geq \lambda\} \begin{cases} \leq \frac{\sigma^2}{\sigma^2 + \lambda^2} & \text{if } \lambda > 0 \\ \geq 1 - \frac{\sigma^2}{\sigma^2 + \lambda^2} & \text{if } \lambda < 0 \end{cases}$$

For male, $E[X] = 83$, $\lambda = 100 - 83 = 17$, $\sigma = 5$

$$P\{X \geq 100\} \leq \frac{25}{25 + 289} \approx 0.079617834 \approx 8.0\%$$

For female, $E[X] = 87.7$, $\lambda = 100 - 87.7 = 12.3$, $\sigma = 5$

$$P\{X \geq 100\} \leq \frac{25}{25 + 151.29} \approx 0.141811787 \approx 14.2\%$$

Comment: This is more reasonable. Note that they are upper bounds. Thus, the actual probability is smaller.

d)

The Markov's inequality can only provide very rough estimates. When the variance is also available, the quality of the estimate is better.

One can get the statistics of the mean death age and variance of the death age at different years. Then use regression to estimate the expected life at birth. We shall study the topic of regression later.

e)

On average, female lives longer than male. As the mean age of Hong Kong population is increasing, the weighting of older people gets higher and higher. Thus, the sex ratio is dropping.

Example 9

100 samples are taken from an i.i.d. random process. (i.i.d. stands for independent and identically distributed). It means that the samples are taken independently and taking a sample will not affect other samples, for example, the samples are taken with replacement randomly. It also means that the samples are taken from the same (i.e., identical) probability distribution.

What is the lower probability bound if it is required that the sample mean and the true mean differs by less than 2 standard deviations?

Caution: Note that you should not assume that the distribution is Gaussian, as the question does not say so. To test whether a distribution is likely to be Gaussian, you need to conduct statistical tests such as normality test.

Answers

The weak law of large numbers can be used to bound the difference between the sample mean and the true mean.

$$P \left\{ \left| \frac{X_1 + \dots + X_n}{n} - \mu \right| \geq \epsilon \right\} \leq \frac{\sigma^2}{n\epsilon^2}$$

Thus,

$$P \left\{ \left| \frac{X_1 + \dots + X_n}{n} - \mu \right| < \epsilon \right\} > 1 - \frac{\sigma^2}{n\epsilon^2}$$

It is required that the difference between the sample mean and the true mean is less than 2 standard deviation. Thus,

$$\epsilon = 2\sigma$$

The lower probability bound is

$$1 - \frac{\sigma^2}{(100)(4\sigma^2)} = 0.9975$$

It means that no matter what the distribution is, the probability is at least 99.75%.

Example 10

Suppose a sample of 100 individuals in a large population is taken and the sample mean age is 40 years. Assume the age in the population is i.i.d. and the minimum age is 0 years and the maximum age is 120 years. What is the probability that the true mean lies between 30 to 50 years?

Use the Hoeffding's inequality.

(Hoeffding's Inequality) Let $\theta_1, \dots, \theta_m$ be a sequence of i.i.d. random variables and assume that for all i , $E[\theta_i] = \mu$ and $P[a \leq \theta_i \leq b] = 1$. Then, for any $\epsilon > 0$,

$$P\left[\left|\frac{1}{m}\sum_{i=1}^m \theta_i - \mu\right| > \epsilon\right] \leq 2\exp(-2m\epsilon^2/(b-a)^2)$$

Answers

$\epsilon = 10$, $m = 100$, $a = 0$, $b = 120$, the probability that the true mean lies between 30 and 50 is larger than

$$1 - 2\exp\left(-\frac{2m\epsilon^2}{(b-a)^2}\right) = 1 - 0.498704417 = 0.501295583$$

The beauty of this inequality is that we do not need to know the distribution and we do not need to know the standard deviation either! We just need to know the minimum and the maximum.